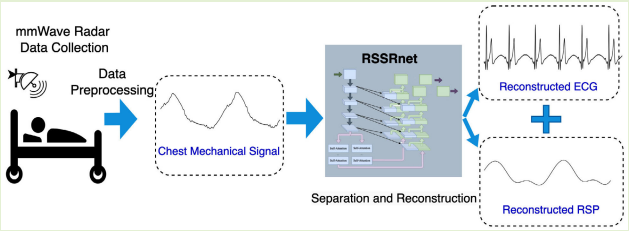
Contactless Reconstruction of ECG and Respiration Signals With mmWave Radar Based on RSSRnet

Yingxiao Wu, *Member, IEEE*, Haocheng Ni, Changlin Mao, and Jianping Han

***Abstract*—In response to the challenge of effectively rebuilding vital signals acquired from radar, we propose RSSRnet that works on the short-time Fourier transform (STFT) spectrum to separate radar-acquired chest mechanical signals and reconstruct electrocardiogram (ECG) signals and respiration (RSP) signals. RSSRnet is constructed based on an encoder–separator–decoder architecture, which combines multiple self-attention blocks to enhance the separated representation of bottleneck features. In addition, we incorporate channel attention (CA) and cross CA (CCA) blocks to refine the feature representation of the encoder and the decoder, respectively. RSSRnet is optimized using an improved loss function defined on both the time-domain waveform and STFT spectrum. The experimental results show that RSSRnet can not only separate chest mechanical signals with mixed respiratory and heartbeat components, but also effectively perform the transformation from mechanical signals to electrical signals. Compared to the ground-truth signals, the median-normalized root mean square error (NRMSE) for reconstructing RSP and ECG signals is 0.043 and 0.049, respectively. The median Pearson correlation coefficients (PCCs) for reconstructed RSP and ECG signals are 0.992 and 0.964, respectively. Furthermore, the median absolute error of the cardiac intervals between the reconstructed ECG signals and the ground-truth values is less than 8 ms.**



***Index Terms*— Deep learning, electrocardiography, radar, respiratory signal, vital signs.**

I. Introduction

VITAL signs monitoring entails the continuous monitoring and analysis of basic physiological parameters of the human body, and it has a wide range of applications. Emotion recognition is a noteworthy application. It infers a person's emotional state by examining vital signs such as heart rate, variability, and respiratory rate [1]. Furthermore, vital signs monitoring finds utility within the cockpit environment, wherein the continuous assessment of a driver's vital signs enables the analysis and quantification of fatigue levels, thereby significantly enhancing overall driving safety [2]. Moreover, vital signals can also be used for identity verification. By utilizing the unique heart signal signature exhibited between individuals, a unique and powerful means of personal identification can thus be established [3].

Among the multitude of signals employed in the biomedical domain, electrocardiogram (ECG) and respiratory signals reign as two of the most prevalent ones. ECG signals furnish information on the heart's electrical activity, captured through the recording of minute potential changes across various regions of the myocardium. Typically, a complete heartbeat cycle of ECG comprises distinctive waveforms, such as the P wave, QRS complex, and T wave [4]. ECG signals find wide-ranging applications in the diagnosis and treatment of heart diseases, including arrhythmia [5], ischemic heart disease [6], and cardiomyopathy [7], [8]. Respiratory signals encapsulate the state of the human respiratory system, characterized by inhalation and exhalation. Monitoring respiratory signals aids in the diagnosis of respiratory ailments such as asthma [9], airflow obstruction [10], and sleep apnea syndrome [11]. The chest strap method represents a commonly employed technique for respiratory signal monitoring, wherein a chest strap affixed to the chest or abdomen detects respiratory signals by measuring changes in strap tension. Another approach is the impedance-based method, which gauges impedance variations resulting from volume changes in the abdomen and chest during breathing.

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However, conventional methods for vital signs monitoring, such as electrocardiography, chest bands, and impedance-based techniques, are beset with numerous limitations. These methods are unsuitable for prolonged monitoring and necessitate physical contact with the human body, thus proving inconvenient for patients. Furthermore, traditional multivital signs signal acquisition methods rely on multiple sensors and data channels for signal acquisition and processing, resulting in heightened system complexity, elevated costs, and susceptibility to interference. Consequently, there is an urgent need for a robust noncontact method capable of monitoring diverse human vital signs.

One significant challenge in utilizing radar as a noncontact device for vital signs detection is the modulation of the wireless signal by both the subject's breathing and heartbeat, making it difficult to separate the two. Additionally, the small amplitude of chest motion caused by breathing and heartbeat makes it susceptible to interference from background noise. Furthermore, the frequencies of respiratory and heartbeat signals are closely aligned, making them prone to mutual interference. For the separation of respiratory and heartbeat signals, bandpass filters (BPFs) are the most rudimentary method [12], [13], [14]. But there is a problem of spectrum leakage, even though the frequency bands of heartbeat and respiration (RSP) do not overlap [15]. In [16], the Kalman filter was used for heart rate tracking, and the authors used the predicted values and uncertainties to further update and narrow the BPF. While this approach improves the accuracy of heart rate estimation, it is unable to extract fine-grained cardiac events. Feng and Pan [17] used empirical modal decomposition (EMD) to adaptively decompose the radar signal into intrinsic mode functions (IMFs). The breathing and heartbeat signals are reconstructed in the time domain by calculating the energy spectral characteristics of each IMF. Nguyen et al. [18] and Yang et al. [19] used ensemble empirical mode decomposition (EEMD) to reduce modal aliasing. However, the introduced noise affects the result accuracy. Variational modal decomposition (VMD) is introduced to vital signs signal processing in [20]. VMD uses a variational structure instead of recursion to improve the decomposition efficiency. However, one drawback of VMD is that it requires setting the number of decomposition layers and penalty factors, which can lead to problems such as inadequate decomposition or overdecomposition.

To extract fine-grained physiological activity events that are essential for health monitoring and disease diagnosis, the conventional algorithms mentioned above are not enough. Some works attempted to convert radar-reflected signals into signals with physiological activity events, such as SCG and ECG. SCG is a mechanical signal that records the heart's vibrations using accelerometers near its apex. It is morphologically closer to radar-detected vital signals than ECG [21]. ECG is an electrical signal that records action potentials in the heart's conduction system using 12-lead electrodes. RF-SCG [22] utilized a 1-D CNN to learn the mapping function from RF signals to SCG-like signals, which allowed for the reconstruction of SCG-like waveforms with high accuracy. The experiments demonstrated that the correlation coefficient of SCG-like waveforms reconstructed from RF-SCG was above 0.72, and the median error for timing five key cardiac events ranged from 0.26% to 1.29%. In [23], ECG-like signals were reconstructed without physical contact. Their study was based on the fact that both the mechanical conduction signal of the heart acquired by radar and the electrical conduction signal, that is, ECG, originate from cardiac activity. Their experiments demonstrated that the median error of the reconstructed ECG waveform was less than 14 ms, the median Pearson correlation coefficient was 90%, and the median root mean square error was 0.081 mv. However, while these models can reconstruct signals with fine-grained cardiac events, they are not able to simultaneously monitor the respiratory signal. Additionally, these models are based on time-domain waveforms that are highly influenced by noise and fluctuations.

In this study, we propose a novel end-to-end deep-learning model, RSSRnet, for the contactless reconstruction of ECG and RSP signals with mmWave radar. RSSRnet can not only separate chest displacement signals with mixed respiratory and heartbeat components, but also effectively perform the transformation from mechanical signals to electrical signals. RSSRnet works on the short-time Fourier transform (STFT) spectrum for separating vital signals acquired by radar and reconstructing both ECG and RSP signals. RSSRnet is constructed based on an encoder–separator–decoder architecture, which combines multiple self-attention blocks to enhance the separated representation of bottleneck features. In addition, we incorporate cross-channel attention (CCA) blocks to refine the feature representation of the encoder and the decoder, respectively. RSSRnet is optimized using an improved loss function defined on both the waveform and STFT spectrum.

The main contribution of this article can be summarized as follows.

1) We build a new method to solve several technical challenges and reconstruct ECG and RSP signals by radar-based sensing. To the best of our knowledge, this study is the first successful effort to accomplish this task.
2) We propose an advanced and high-performance deep-learning network, called RSSRnet, for disentangling the heartbeat and respiratory components mixed in the chest displacement signal and establishing an accurate mapping from mechanical signals to electrical signals.
3) We novelly employ STFT to emphasize the patterns of heartbeat and respiration mixed in the radar-acquired chest mechanical signal. We propose a CCA module and improved loss functions defined in both the time domain and STFT spectrum within the proposed RSSRnet. Ablation experiments are conducted to demonstrate the effectiveness of these enhancements in improving overall performance.
4) Experimental results showed the outstanding performance of RSSRnet. Compared to the ground-truth signals, the median normalized root mean square error (NRMSE) for reconstructed RSP and ECG signals are 0.043 and 0.049, respectively. The median Pearson correlation coefficients (PCCs) for reconstructed RSP

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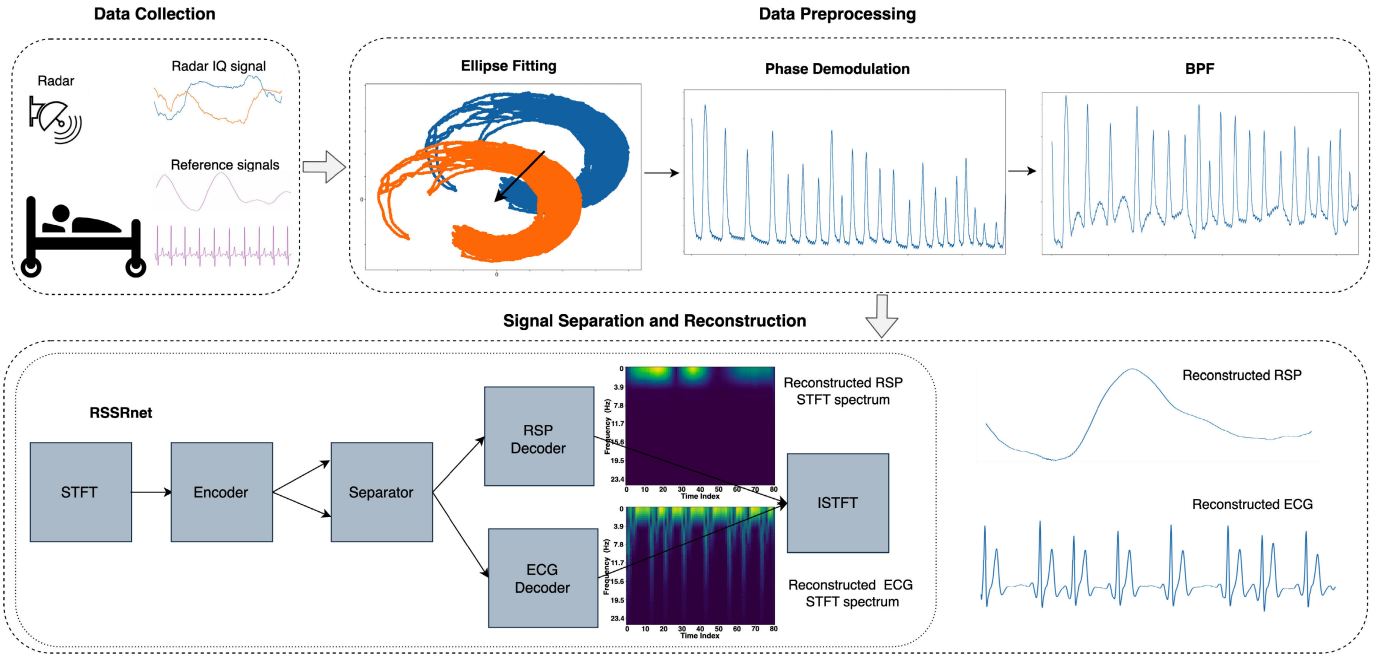


Fig. 1. System overview.

and ECG signals are 0.992 and 0.964, respectively. Furthermore, the median absolute error of the cardiac intervals between the reconstructed ECG signals and the ground-truth values is less than 8 ms.

II. System Overview and Data Processing

A. System Overview

Fig. 1 provides a comprehensive system overview of the proposed method, highlighting the key steps involved.

The process begins with data acquisition to collect the mmWave radar signal reflection from the human body, alongside the simultaneous recording of the ECG and RSP signals as ground truth.

To enhance the quality of the radar reflection signals, a series of preprocessing techniques are employed. Initially, an ellipse fitting approach is utilized to solve dc offset. Subsequently, the signals underwent a differential and cross-multiplication (DACM) algorithm, which is used for phase demodulation. Furthermore, a BPF is used for removing the frequency components of the nonvital signal to obtain the chest mechanical signal.

Finally, RSSRnet, a deep-learning model with an encoder–separator–decoder structure, is harnessed to segregate chest mechanical signals encompassing mixed respiratory and heartbeat components and transform them into electrical signals. Specifically, as depicted in Fig. 1, the RSP decoder and the ECG decoder of RSSRnet accomplish the reconstruction of the STFT spectrum of RSP and ECG, respectively, subsequently yielding the reconstructed RSP and ECG signals via inverse STFT (ISTFT).

B. Data Preprocessing

The radar-collected signal is mixed with chest displacement signals of heartbeat and respiration, noise, and other clutter. This section elaborates on radar data preprocessing techniques to obtain chest displacement signals mixed with heartbeat and respiratory components. We describe the processing method applied to the simultaneously acquired ECG and RSP signals.

The acquisition of radar signals is performed using a 24-GHz continuous wave radar system based on the six-port technology [24]. This radar system comprises two input ports and four output ports, where the input signals are combined with four predetermined and fixed phase shifts. Subsequently, the resulting output signals are downconverted to baseband signals using diode power detectors. Consequently, the obtained baseband signals accurately represent the orthogonal in-phase (I) and quadrature (Q) components of a complex number. The phase change $\Delta\Phi$ between the transmitted (Tx) and received (Rx) signals, along with the chest motion Δx induced by the respiratory and cardiac activity of the target, can be mathematically described as follows:

$$\Delta\Phi = 2\pi \frac{\Delta x}{\lambda} \tag{1}$$

where λ is the wavelength of the RF signals.

However, the IQ signal in practical hardware implementations often suffers from imperfections, such as the amplitude or phase mismatch. Hence, before phase demodulation, the calibration of the IQ signals is necessary. Ideally, when the IQ signal is perfectly matched, the arc trajectory in the complex domain conforms to a circular shape centered at the origin. Conversely, if the IQ signal is mismatched, the arc trajectory will deviate from the origin and exhibit the form of an ellipse. Consequently, ellipse fitting becomes crucial for calibration, and various techniques can be employed for this purpose, including least-squares fitting [25], gradient descent [26], and the statistical random sample consensus (RANSAC) algorithm [27]. As shown in Fig. 1, the center of the ellipse is moved

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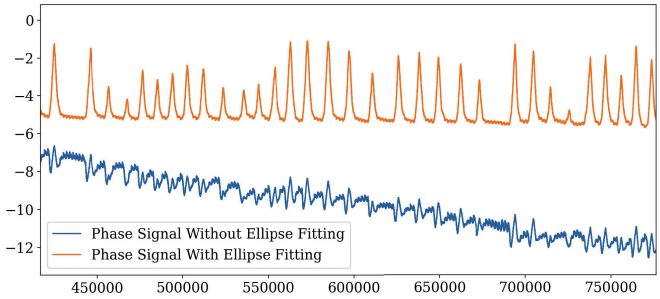


Fig. 2. Comparison of phase signals without ellipse fitting and with ellipse fitting.

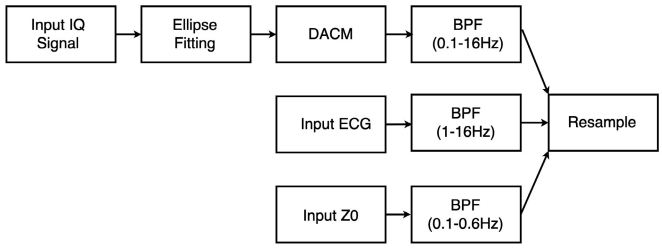


Fig. 3. Flowchart of data preprocessing steps.

to the origin after the ellipse fitting is completed for the IQ signal.

Subsequently, phase demodulation is performed using the DACM algorithm, which employs the derivative of the arctangent function to avoid calculating trigonometric functions. The process, as outlined in [28], can be expressed through the following equation:

$$\Phi[n] = \sum_{k=2}^{n} \frac{I[k]\{Q[k] - Q[k-1]\} - \{I[k] - I[k-1]\}Q[k]}{I[k]^2 + Q[k]^2} \tag{2}$$

where forward differences and cumulants are used to approximate differentiation and integration, respectively. This calculation can solve the phase discontinuity problem that arises in the arctangent demodulation. In Fig. 2, we show the phase signal obtained after ellipse fitting, as well as the phase signal without ellipse fitting. It is evident that the signal without ellipse fitting not only suffers from dc offset, but also exhibits significant distortion. Next, the phase signal is filtered using a fourth-order Butterworth filter with a bandpass range of 0.1–16 Hz, to remove nonbreathing and nonheartbeat components.

As shown in Fig. 3, the ECG and the impedance signal are processed using a BPF. Specifically, a fourth-order Butterworth filter with a bandpass range of 1–16 Hz is used to eliminate physiological and electrical noise and extract the informative components of the ECG signal, as suggested in [29]. The impedance signal is processed using a fourth-order Butterworth filter with a bandpass range of 0.1–0.6 Hz [30]. This frequency range aligns with the typical human respiratory rate range of 6–36 breaths/min. Finally, all signals are resampled to a uniform rate of 100 Hz.

III. Model

In this section, we propose RSSRnet for efficiently separating the mixed heartbeat and respiration components of the radar-acquired chest displacement signal and establishing an accurate mapping from the displacement signal to the ECG and RSP signals.

As shown in Fig. 4, RSSRnet is designed in U-Net style and consists of an encoder, a separator, and two decoders. The encoder and decoders are interconnected by skip connections to ensure efficient information flow in the model. The encoder uses convolutional operations to extract time–frequency features of heartbeat and respiration from the STFT spectra of the radar-acquired chest displacement signals and maps these features to dimensions suitable for further processing by the separator. The separator has two branches that receive high-level features from the encoder. The two branches use the self-attention module to focus on heartbeat and respiration features, respectively, and model the long-term dependencies. The two decoders receive heartbeat and respiratory features from the separator and use inverse convolution operations to reconstruct the STFT spectra of the RSP and ECG signals, and then the ECG and RSP signals are obtained by the ISTFT. To refine the feature representations, RSSRnet introduces channel attention (CA) and our proposed CCA module, which is applied to the encoder and the decoder, respectively. Finally, we employ an improved loss function which is defined over the waveform and STFT spectra to efficiently guide the training of the model. Together, these designs and methods constitute the key elements of RSSRnet.

A. Encoder

In this section, we present the structure of the encoder, which is responsible for encoding the vital signals obtained by radar into high-level features. Given a radar-acquired chest displacement signal $x(t)$ that contains the respiratory and cardiac components, the STFT module in the encoder first converts it to the STFT representation $X \in \mathbb{R}^{F\times T}$, where T stands for the timesteps and F is the frequency bins

$$\begin{aligned} X &= \mathrm{STFT}(x(t)) \\ &= A\, e^{jP} \end{aligned} \tag{3}$$

where A and P refer to the amplitude and phase spectra of X, respectively. The amplitude spectra A and phase spectra P are then stacked as the input of the encoder block.

The encoder block consists of 2-D convolution, CA, batch normalization, and parametric rectified linear unit (PReLU) activation function. The encoder block performs a $2\times$ downsampling on the frequency dimension of the feature map and a $2\times$ upsampling on the channel dimension. Thus, the ith encoder layer output can be expressed as $O_i \in \mathbb{R}^{C*2^{i-1}\times(F/2^{i-1})\times T}$, where C denotes the number of filters in the first encoder block.

Similar to [31], the CA module in the encoder block plays a crucial role in feature selection and refinement. As illustrated in Fig. 5, the CA module takes an input feature map $M \in \mathbb{R}^{C\times F\times T}$ and performs several operations to refine

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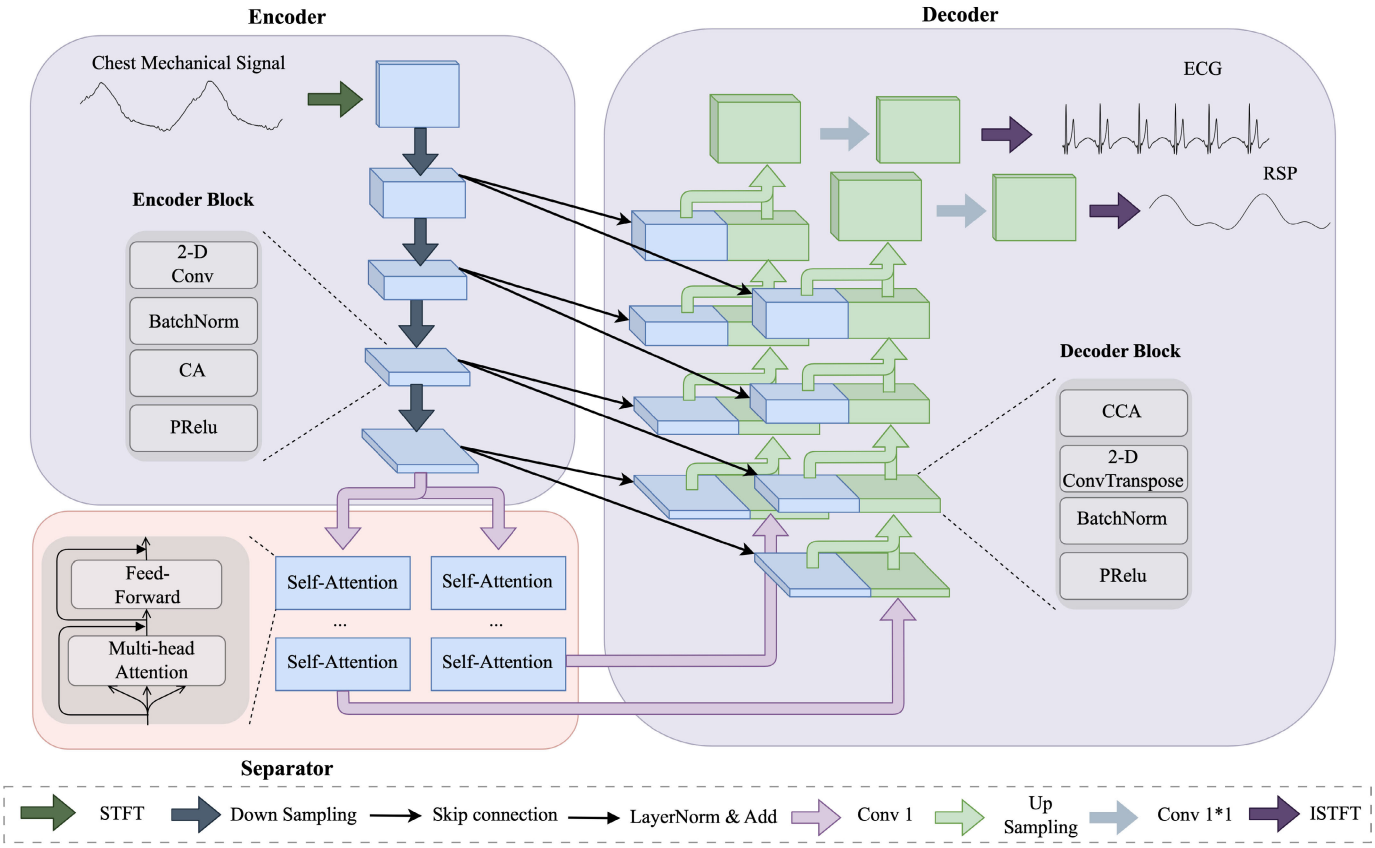


Fig. 4. Overall structure of RSSRnet.

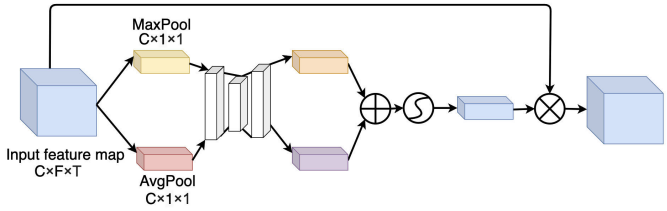


Fig. 5. Architecture of CA.

the features. Initially, the CA module computes the average channel feature $M_{\text{avg}} \in \mathbb{R}^{C\times 1\times 1}$ and max channel feature $M_{\max} \in \mathbb{R}^{C\times 1\times 1}$ using global average pooling and global max-pooling, respectively. Then, the module forwards these two features to two multilayer perceptrons (MLP) with shared weights. After the MLP, the features are transformed by the sigmoid function σ to produce attention weights. Finally, the feature map is calibrated by multiplying the attention weights with the original feature map. The entire process can be represented as follows:

$$M_{\text{att}} = \sigma\big(\text{MLP}\big(M_{\text{avg}}\big) + \text{MLP}(M_{\max})\big) * M. \tag{4}$$

B. Separator

As depicted in Fig. 4, the separator comprises two branches, each of which receives the encoded representation E. These branches are made up of transformer blocks that use self-attention to focus on the respiratory and heartbeat components in H, resulting in effective modeling of long-term dependencies [32].

Initially, the separator applies a 1-D convolution to compress or dilate the encoded representation $O \in \mathbb{R}^{(C\times F)\times T}$

$$H = \text{Conv1D}(O). \tag{5}$$

Here, $H \in \mathbb{R}^{D\times T}$, where D represents the model dimension of the self-attention block.

According to [32], the self-attention block employs multihead attention, where each attention head calculates the scaled dot-product attention between all sequence elements. The result then undergoes layer normalization and residual connection

$$H' = \text{LayerNorm}(\text{MultiHeadAttention}(H)) + H. \tag{6}$$

The output H' then passes through a feedforward network to obtain H''. The feedforward network consists of two MLP and activation functions, layer normalization, and residual connection

$$H'' = \text{LayerNorm}\big(\text{FeedForward}\big(H'\big)\big) + H'. \tag{7}$$

After H passes through two branching structures of the separator stacked by self-attention blocks, we obtain the breathing features $H_r \in \mathbb{R}^{D\times T}$ and heartbeat features $H_e \in \mathbb{R}^{D\times T}$, which are then transformed to the original dimension

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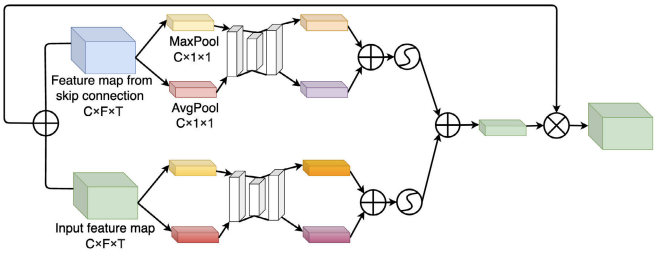


Fig. 6. Architecture of CCA.

using 1-D convolution

$$H_r' = \text{Conv1D}(H_r) \tag{8}$$

$$H_e' = \text{Conv1D}(H_e) \tag{9}$$

where $H_r' \in \mathbb{R}^{(C\times F)\times T}$, and $H_c' \in \mathbb{R}^{(C\times F)\times T}$.

C. Decoder

The architecture of RSSRnet includes two decoders, as illustrated in Fig. 4. These decoders take the output of the separator, H_r' and H_e', and use them to reconstruct the STFT spectrum of the RSP signal and ECG signal, respectively.

To refine the features inside the decoder, we propose the CCA module in the decoder block. The CCA module effectively selects the desired features from the skip connections of the encoder and then fuses them with the input features of the decoder block. As shown in Fig. 4, the decoder block also includes 2-D convolution transpose, batch normalization, and PReLU modules.

As shown in Fig. 6, CCA computes attention weights separately for the feature maps from the skip connections and the decoder block input, following a method similar to that used in CA. The resulting attention weights are added together and multiplied by the sum of the two feature maps to fuse and refine the features.

Unlike the encoder block, the decoder block performs $2\times$ upsampling on the frequency dimension of the feature map and $2\times$ downsampling on the channel dimension. As a result, the ith layer outputs of the respiratory decoder and the ECG decoder, denoted as D_{ri} and D_{ei}, respectively, can be expressed as $D_{ri}/D_{ei} \in \mathbb{R}^{C*2^{i-1}\times(F/2^{i-1})\times T}$. Finally, a 1×1 2-D convolution is used to compress the number of channels to 2, indicating the real and imaginary components of the STFT spectrum. These components are subsequently combined to construct the STFT spectra of both the ECG signal $E \in \mathbb{R}^{F\times T}$ and the impedance respiratory signal $R \in \mathbb{R}^{F\times T}$.

The ECG waveform, e, and respiratory waveform, r, can be obtained by applying the ISTFT to the STFT spectra, as follows:

$$e = \text{ISTFT}(E) \tag{10}$$

$$r = \text{ISTFT}(R). \tag{11}$$

D. Loss Function

To enhance the efficacy of feature extraction and optimize model performance, we propose a novel loss function defined on the time-domain waveform and STFT spectrum. The loss function is formulated as

$$L = \alpha_1 \times L_{\text{wave}} + \alpha_2 \times L_{\text{spectral}} \tag{12}$$

where α_1 and α_2 denote the weights assigned to L_{wave} and L_{spectral}, respectively. The first component, L_{wave}, is defined as

$$L_{\text{wave}} = \frac{1}{N}\sum_{i=0}^{N-1}\left|e_i - \hat{e}_i\right| \tag{13}$$

where e, $\hat{e}$, and N represent the reconstructed ECG sample, the ground-truth ECG sample, and the total number of samples in the waveform, respectively.

The second component, L_{spectral}, is defined as

$$L_{\text{spectral}} = \frac{1}{TF}\sum_{t=0}^{T-1}\sum_{f=0}^{F-1}\left|\left(|E(t,f)| - \left|\hat{E}(t,f)\right|\right)\right| \tag{14}$$

where $|E|$ and $|\hat{E}|$ denote the magnitude spectra of the ECG and the ground truth, respectively. T and F represent the number of frames and frequency bins, respectively. The same loss function is also used for respiratory signals.

IV. Experiment and Results

A. Dataset Description

The dataset utilized in this study is publicly available and provided by Schellenberger et al. [33]. It was obtained through a meticulously conducted data collection process under controlled conditions by healthcare professionals from the Department of Palliative Medicine at the University Hospital Erlangen. A total of 30 participants were recruited for the study. The dataset encompasses five distinct scenarios, namely Resting, Valsalva, Apnea, Tilt up, and Tilt down. As depicted in Fig. 7, the subject is lying on a tilted table during the experiment. A 24-GHz continuous wave radar is mounted approximately 40 cm above the skin of the subject's chest. The inclination angle of the antenna beams is $\pm 10°$ for transmitting and receiving antennas, respectively, with a focal point at 40 cm. The two raw signal components I and Q are digitized simultaneously using a 24-bit analog-to-digital converter with a sampling frequency of 2000 Hz.

In addition to radar systems, Schellenberger et al. [33] used a reference system to record a variety of contact signals. Among these, ECG and RSP signals are recorded at sampling rates of 1000 and 100 Hz, respectively. The technical details of the synchronization of the radar system and the reference system are described in detail. We select well-labeled data from 25 subjects in the resting scenario for the experiment. In the preprocessing stage, the data samples within each data pair are resampled to 100 Hz and segmented into 8-s lengths, with a sampling step set at 20% of that length. Furthermore, to maintain consistent and comparable data ranges, a min-max normalization method is applied to normalize each data pair.

B. Implement Details

To evaluate the performance of our model, we conducted a fivefold cross-validation experiment. In each fold, 80% of the data was used for training, while the remaining 20% was

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Fig. 7. Data acquisition environment [33].

held out for testing. We repeated this process five times to ensure that each portion of the data was independently tested at least once. We then average the results over all folds to obtain an overall performance measure. To prevent overfitting of the model, we implement an early stopping strategy based on the validation data metrics, which helps to ensure its generalization ability [34].

All experiments are conducted on a server equipped with an Intel i7-12700 processor and Nvidia 3080 (12 GB). We implement the model using PyTorch [35]. Throughout the training process, we use a mini-batch size of 20, a learning rate of 1e-4, and the Adam optimizer with momentum $\beta_1 = 0.9$ and denominator momentum $\beta_2 = 0.999$ to minimize the loss function and update the network parameters [36].

C. Evaluation Metrics

To accurately evaluate the performance of the model, we utilize PCC and NRMSE as metrics to measure the accuracy of the reconstructed ECG and respiratory signals.

PCC is used to measure the degree of covariance between the reconstructed signal and its corresponding ground truth. This coefficient ranges from -1 to 1, where the magnitude indicates the strength of the correlation and the sign determines whether the correlation is positive or negative. The mathematical expression for PCC is given by

$$\mathrm{PCC} = \frac{\left(x' - \overline{x'}\right)^T (x - \overline{x})}{\left\|x' - \overline{x'}\right\|_2 \|x - \overline{x}\|_2} \tag{15}$$

where x represents the reconstructed signal and x' represents the ground-truth signal. Euclidean norm is denoted by $\|*\|_2$, and the mean function is represented by $\overline{*}$.

Another metric we employ is NRMSE, which measures the differences between the values of the reconstructed and ground-truth signals. A smaller value of NRMSE indicates better agreement between the predicted and actual values. The use of normalization in the formula ensures that the NRMSE is independent of the scale of the data. The mathematical expression for NRMSE is given by

$$\mathrm{NRMSE} = \frac{\sqrt{\frac{1}{N}\sum_{i=1}^{N}\left|x_i - x_i'\right|^2}}{x_{\max} - x_{\min}} \tag{16}$$

where N represents the sample size, while $x_{\max}$ and $x_{\min}$ denote the maximum and minimum values of the signal, respectively.

D. Ablation Studies

To fully evaluate the model structure of RSSRnet, we compared it with several variant models. These models include.

1) *RSSRnet:* RSSRnet utilizes STFT and ISTFT with a window length and hop size of 0.4 and 0.1 s, respectively. The FFT length is set to 256, and the first 64 frequency bins are selected as the model input. The feature map channels, denoted as C, are set to 32. The encoder and decoder blocks consist of four layers, with kernel size and stride set to (5, 2) and (2, 1), respectively. Within the self-attention block, the model dimension D is set to 512, the feed-forward dimension is 1024, and the head number is 4. The weights α_1 and α_2 in the loss function are assigned values of 0.9 and 0.1, respectively.
2) *RSSRnet-S:* RSSRnet-S can be considered as a 1-D convolutional version of RSSRnet, which removes the STFT module and directly takes time-domain waveforms as input. The encoder and decoder consist of five layers, with a kernel size and stride set to 4 and 2, respectively.
3) *RSSRnet-C:* RSSRnet-C is a version of RSSRnet without the CCA module.
4) *RSSRnet-L:* RSSRnet-L is a version of RSSRnet with the spectral loss L_{spectral} removed.

Table I presents the results of fivefold cross-validations for RSSRnet and its variants, employing various evaluation metrics. The findings demonstrate the superior performance of RSSRnet in comparison to the other variants across all metrics. Notably, in terms of ECG reconstruction, RSSRnet achieves a remarkably high PCC of 0.959 ± 0.039, indicating a strong resemblance between the reconstructed signals and the ground truth. Moreover, the NRMSE for ECG reconstruction is impressively low at 0.051 ± 0.022, signifying accurate reconstruction with minimal residual error. Similarly, RSSRnet exhibits favorable results for RSP reconstruction, with a PCC of 0.978 ± 0.068 and NRMSE of 0.054 ± 0.044.

In contrast, RSSRnet-S, which directly processes the time-domain waveform without employing the STFT module, exhibits inadequate performance in ECG signal reconstruction. Although the reconstruction of the RSP signal remains acceptable, the PCC and NRMSE values for ECG reconstruction are significantly lower, measuring 0.008 and 0.205, respectively. These outcomes suggest that the utilization of the 1-D time-domain convolution solely is insufficient in effectively capturing the weaker components of the heartbeat within the mixed waveform.

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TABLE I
RESULTS OF FIVEFOLD CROSS-VALIDATIONS OF RSSRNET AND ITS VARIANTS. THE PCC AND NRMSE ARE IN "MEAN ± STD" FORMAT

Model	Param (M)	ECG		RSP	
		PCC	NRMSE	PCC	NRMSE
RSSRnet	17.1	0.959±0.039	0.051±0.022	0.978±0.068	0.054±0.044
RSSRnet-S	17.5	0.008±0.044	0.205±0.040	0.895±0.197	0.151±0.058
RSSRnet-C	16.7	0.948±0.045	0.062±0.029	0.960±0.140	0.086±0.058
RSSRnet-L	17.1	0.951±0.038	0.051±0.023	0.976±0.080	0.058±0.047

Furthermore, the performance of RSSRnet-C, the variant lacking the CCA module, demonstrates a decline in comparison to RSSRnet. The PCC value for ECG reconstruction slightly decreases to 0.948 ± 0.045, while the NRMSE slightly increases to 0.062 ± 0.029. As for RSP reconstruction, RSSRnet-C achieves a PCC value of 0.960 ± 0.140 and NRMSE of 0.086 ± 0.058. These results indicate that the CCA module effectively captures the distinct components of the heartbeat and respiration present in the mixed signals, thereby enhancing the reconstruction performance.

Regarding RSSRnet-L, which excludes spectral loss, a minor deterioration in performance is observed for both ECG and RSP reconstructions. Notably, RSSRnet with spectral loss yields superior outcomes, exemplified in Fig. 8, by successfully recovering ECG peaks and generating smoother RSP signals. These results underscore the efficacy of the spectral loss in ensuring stable model training and improving the accuracy of waveform reconstruction.

Based on the comprehensive comparison of the different variant models, it is determined that the STFT module, CCA module, and spectral loss are effective. As shown in Table I, RSSRnet has a parameter count close to that of the other variants of the model, but RSSRnet efficiently captures the characteristics of heartbeat and respiration within the mixed signals, accurately reconstructing the ECG and RSP signals. The computational complexity of RSSRnet is low, typically about 2.07 Giga FLoating-number OPerations for one action sample.

E. Accuracy of Signal Reconstruction

First, we assess the morphology of the reconstructed signals. In Fig. 9(a) and (b), the cumulative distribution functions (CDFs) of NRMSE and PCC for ECG and RSP signals are plotted. Observing the results, it is evident that more than 90% of the NRMSE values are below 0.1, indicating a high level of accuracy. Additionally, over 90% of the PCC values exceed 0.9, further affirming the fidelity of the reconstructed signals. Fig. 9(c) and (d) presents the median NRMSE and PCC between the reconstructed ECG and RSP signals and their corresponding ground truth. The median NRMSE for the reconstructed ECG signal is 0.049, while the median PCC is 0.964. For the reconstructed RSP signal, the median NRMSE is 0.043, and the median PCC is 0.992.

These findings provide compelling evidence that RSSRnet exhibits reliable and accurate performance in separating the cardiac and respiratory components from mixed waveforms and successfully reconstructing ECG and RSP signals.

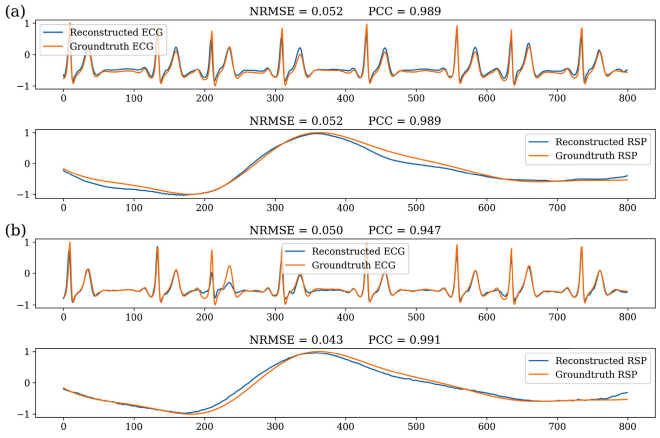


Fig. 8. Qualitative comparison of waveform reconstruction. Example output from (a) RSSRnet and (b) RSSRnet-L.

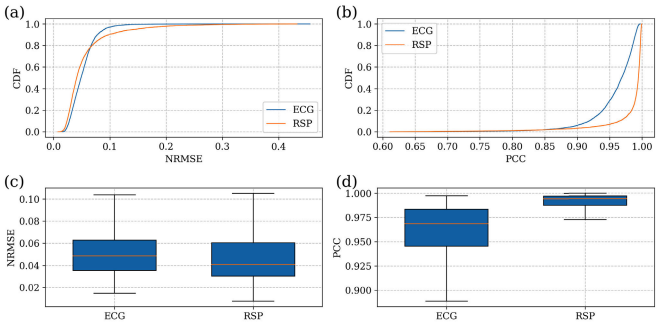


Fig. 9. Accuracy of signal morphology. (a) CDFs of the overall NRMSE for reconstructed ECG and RSP signals, (b) CDFs of the overall PCC for reconstructed ECG and RSP signals, (c) NRMSE of ECG and RSP to ground truth, and (d) PCC of ECG and RSP to ground truth.

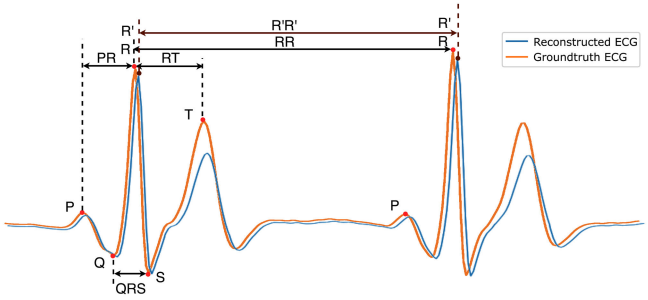


Fig. 10. Diagram of cardiac intervals.

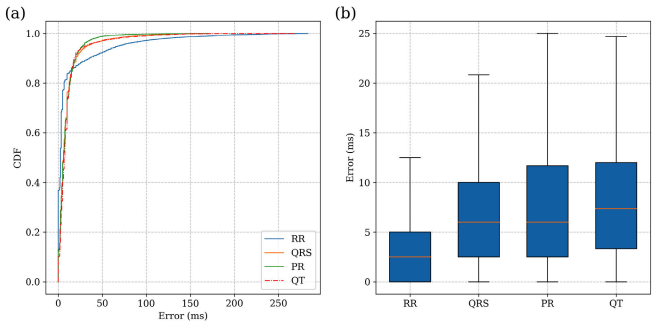


Fig. 11. Accuracy of ECG intervals. (a) CDFs of the overall absolute timing error for cardiac interval timing and (b) absolute timing error of cardiac interval timing.

Furthermore, the accuracy of the reconstructed ECG signals is evaluated in terms of cardiac interval timing, encompassing

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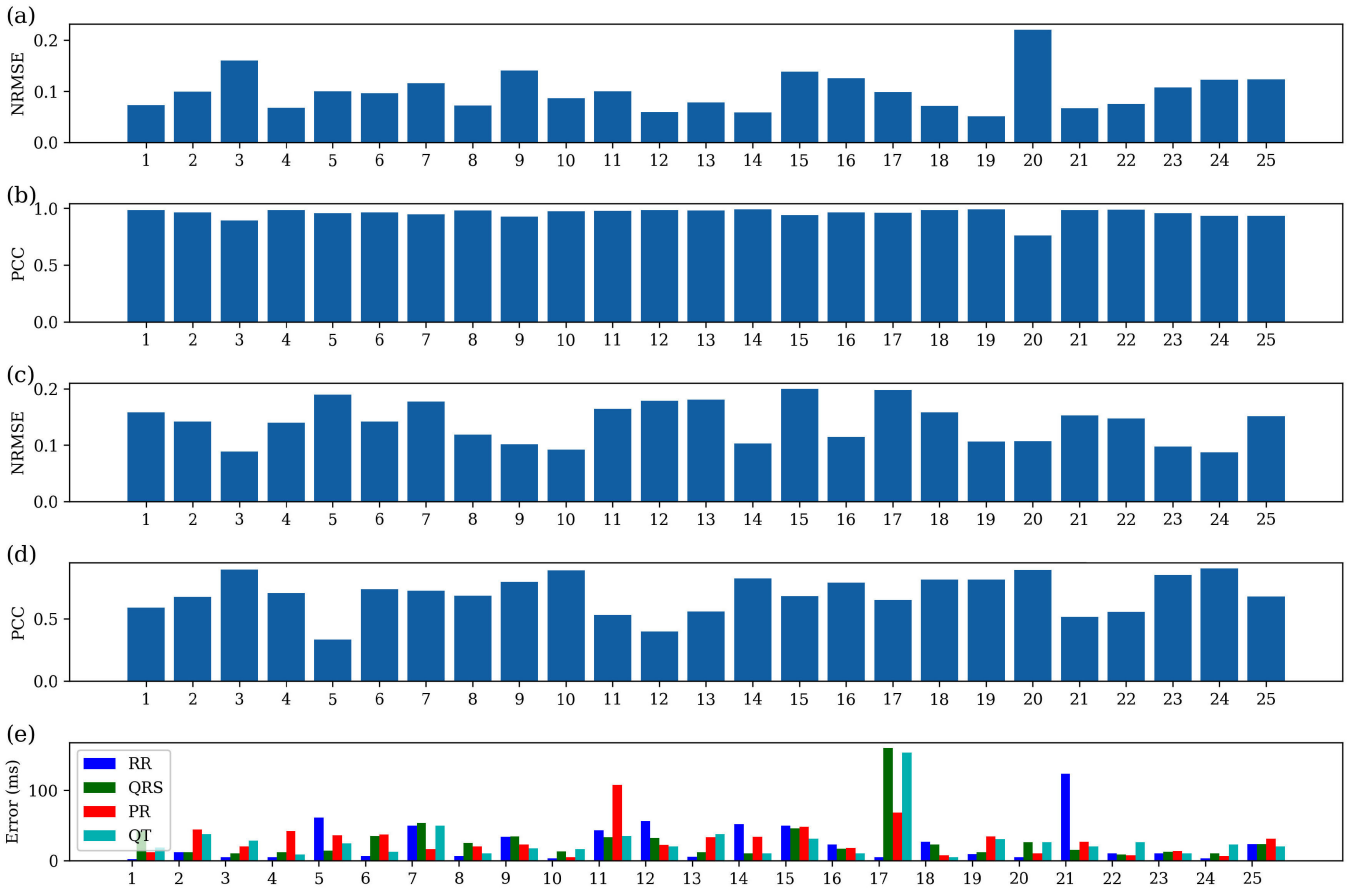


Fig. 12. Results across subjects. (a) Median NRMSE of the reconstructed RSP to ground truth across subjects, (b) median PCC of the reconstructed RSP to ground truth across subjects, (c) median NRMSE of the reconstructed ECG to ground truth across subjects, (d) median PCC of the reconstructed ECG to ground truth across subjects, and (e) median absolute timing error of the cardiac intervals across subjects.

the RR interval, QRS interval, PR interval, and RT interval, as illustrated in Fig. 10. These intervals serve as crucial reference indicators for disease diagnosis in clinical practice. To analyze the intervals, NeuroKit2 [37], an open-source physiological signal-processing tool, is utilized to detect the peaks of the ECG signal and calculate the durations of the intervals. The absolute errors between the reconstructed ECG intervals and the intervals obtained from the ground-truth ECG are employed to assess the accuracy of cardiac interval timing in the reconstructed ECG. As illustrated in Fig. 11, more than 90% of the errors for each interval are below 50 ms. After removing outliers, the median errors for the RR interval, QRS interval, PR interval, and RT interval are measured to be 2.5, 6.0, 6.0, and 7.3 ms, respectively.

These results further validate the accuracy of RSSRnet in reconstructing ECG signals and its ability to capture cardiac interval timing, which is crucial for clinical diagnosis and analysis.

F. Results Across Subjects

In this section, the accuracy of signal reconstruction by RSSRnet is evaluated.

To evaluate the generalization performance of the RSSRnet model, a rigorous 25-fold cross-validation strategy is employed. The dataset is partitioned into 25 subsets, with the model trained on data from 24 individuals and evaluated on the remaining one. The evaluation results, depicted in Fig. 12, illustrate the median values of 25 subject-wise evaluation metrics.

As shown in Fig. 12(a) and (b), the NRMSE of reconstructed RSP signals is between 0.05 and 0.22, indicating the high quality of the reconstruction. Moreover, the PCC, ranging from 0.76 to 0.99, further validates the accuracy of the reconstructed signals. Impressively, 17 individuals achieved an NRMSE below 0.1, while 24 individuals exhibited a PCC above 0.9.

Turning to the reconstructed ECG signals shown in Fig. 12(c) and (d), similar performance assessments are conducted. The NRMSE values ranged from 0.08 to 0.20, reflecting the fidelity of the reconstruction process. Additionally, the PCC index spanned from 0.39 to 0.90, elucidating the similarity between the original and reconstructed signals. Encouragingly, more than half of the individuals demonstrate a PCC above 0.7, underscoring the reliability of the reconstruction results.

Fig. 12(e) shows the error of cardiac intervals for each participant. Notably, the observed maximum error for RR intervals is 123 ms, while the remaining RR interval errors are all below 60 ms, with a substantial proportion below 10 ms. For QRS intervals, the maximum error reaches 160 ms, with

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the remaining errors below 50 ms. Similarly, the maximum errors for PR and QT intervals are 108 and 153 ms, respectively, with the majority of interval errors falling below 50 ms. These findings further affirm the accuracy and robustness of the proposed RSSRnet model in accurately estimating cardiac intervals.

The results obtained through the 25-fold cross-validation experiment demonstrate the generalization capabilities of the RSSRnet model. The reconstructed RSP and ECG signals exhibit favorable NRMSE and PCC, while the individual errors associated with cardiac intervals are generally within acceptable limits, highlighting the efficacy of the proposed approach in accurately capturing physiological signals.

V. Conclusion

This article presents a new method for separating and reconstructing ECG and RSP signals from radar-acquired chest displacement signals. We novelly use the STFT to highlight the heartbeat and respiration patterns mixed in the chest displacement signals acquired by radar. RSSRnet is proposed to efficiently untangle these intertwined components and establish an accurate mapping from mechanical signals to electrical signals. Notably, we developed a CCA module and an optimized loss function in the RSSRnet model, which significantly improved the performance of signal separation and reconstruction. The median NRMSE and PCC values of the reconstructed RSP signal relative to the ground truth are 0.043 and 0.992, respectively. The median NRMSE and PCC values of the reconstructed ECG signal relative to the ground truth are 0.049 and 0.964, respectively. This study is expected to advance the development of vital sign-sensing technology and the application of noninvasive health monitoring.

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